



Office of Science Graduate Student Research

2025 Solicitation 1 - Final Report for: Jacob Hauck

OVERVIEW

Awardee and Project Information

Awardee Name:	Jacob Hauck
Graduate Institution:	Missouri University of Science & Technology
Discipline:	Mathematics
Doctoral Dissertation/Thesis Title:	Deep Learning--Based Surrogate Models for Nonlocal PDEs with Applications to Fracture Dynamics
Degree Objective:	Ph.D.
Expected Graduation Date:	05/2027
SCGSR Research Project Title:	Learning Surrogate Models For Nonlocal Mechanics Problems
SCGSR Research Project Abstract:	Traditional methods for numerical simulations encounter significant computational challenges when applied to nonlocal problems, particularly at large scales. Recent work on operator learning, including a model that we have developed, in which one uses machine learning to fit a surrogate solver to data, has shown promise in overcoming these challenges. Despite strong successes on local problems, operator learning for nonlocal problems remains largely unexplored because the lack of high-quality datasets and benchmarks for nonlocal problems has impeded effective training, validation, and algorithm design. In this project we will focus on generalizing the operator learning model that we have already developed to mechanics problems based on peridynamics, which are an important and difficult class of nonlocal problems. We will establish benchmark examples and generate corresponding datasets using PDMATLAB2D and CabanaPD software. With these datasets we will develop and test operator learning methods tailored to fracture dynamics. As a result, this work will contribute significantly to my Ph.D. dissertation and to the operator learning and nonlocal communities more broadly.
Starting Date:	3/2/2026
Ending Date:	9/4/2026
Host DOE Laboratory:	Oak Ridge National Laboratory (ORNL)
Division at the Host DOE Laboratory:	Computer Science and Mathematics
Collaborating DOE Laboratory Scientist Name:	Pablo Seleson
Collaborating DOE Laboratory Scientist Email:	selesonpd@ornl.gov
Supervisor for Collaborating DOE Laboratory Scientist:	N/A
Scientific user facility involved in the SCGSR research project:	Oak Ridge National Laboratory Leadership Computing Facility (OLF)

ACCOMPLISHMENTS

Research Goals and Objectives

Research Goals and Objectives:	As stated in the proposal, the main goal of this project is the development of operator learning algorithms for surrogate modeling in nonlocal solid mechanics. This overall goal is broken down into two scientific objectives: 1. Generalizing our prior work on operator learning to be applicable to problems with vector-valued functions (which are typical in mechanics). 2. Developing operator learning techniques capable of handling discontinuities that arise in fracture dynamics. We proposed a two-part technical approach to achieve each of these objectives. In the first part we will develop the modifications needed to adapt our operator learning methods to the challenge of the objective. In the second part we will develop a benchmark dataset that can be used to test the effectiveness of the ideas developed in the first part. In the case of generalizing our method to vector-values functions, we first proposed to modify the architecture of the neural networks used in our method so that they can represent vector-valued functions. This will require minor changes to the mathematical theory and code implementation (which is written in PyTorch). Second, we proposed to develop a benchmark dataset for wave propagation, in which one uses a surrogate model to approximate the mapping from the initial state of the displacement field to final state (i.e.,
---------------------------------------	---

approximate the propagation of displacement waves). The dataset is to be generated by running wave propagation simulations with the PDMATLAB2D software using random initial states sampled from a Gaussian random field.

In the case of developing operator learning techniques for problems with discontinuities, we proposed to compare linear and nonlinear reconstruction methods for representing output functions, as this has been identified in the literature as a key factor contributing to accuracy when functions with discontinuities are present. In addition, since operator learning for problems with discontinuities and for fracture dynamics, in particular, is a relatively new area of study, we also proposed to assess the effectiveness of existing techniques and to identify any new challenges. Finally, we proposed to develop and generate a novel dataset for crack branching, which is a challenging problem in fracture dynamics. The dataset is to be generated using the PDMATLAB2D software for 2D problems and the CabanaPD software for 3D problems.

Results, Discussion, and Conclusions/Accomplishments

Project Accomplishments:

Results and Outcomes

This project is split into two parts: operator learning for wave propagation and for crack propagation. We obtained more results for wave propagation due to challenges with sensitivity for crack propagation, discussed in more detail below; however, we achieved significant outcomes in both parts.

Wave propagation

In this part we applied our operator learning model to wave propagation problems. Here, the training data consists of pairs of input and output functions that are the initial and final displacement fields of a wave propagation simulation. We then train our operator learning model to predict the final displacement field directly from the initial displacement field.

Outcomes and conclusions

We needed to reformulate our operator learning architecture, depicted in Figure 1, to model the vector-valued displacement field that appears in mechanics. To this end, we used vector-valued neural network basis functions in both the encoder and reconstructor modules. The resulting model is described in Equation (1).

Next, we generated benchmark datasets for wave propagation using the software PDMATLAB2D. Figure 2 presents sample input and output functions. We generated several specialized datasets to explore potential difficulties for operator learning. We generated datasets of varying complexity, as shown in Figure 3, to evaluate how well operator learning models work in more complex scenarios, and we generated datasets with different simulation durations to evaluate how operator learning performance depends on simulation time. Figure 4 shows examples of data we generated to test if operator learning models can generalize well to out-of-distribution data, and Figure 5 shows examples of data we generated using an anisotropic material model, which required modifying PDMATLAB2D to enable anisotropic simulations.

We then evaluated the effectiveness of our operator learning model at predicting wave propagation behavior. We started by exploring the effects of model hyperparameters, as outlined in Table 1, finding that a deep approximator network produced better results than a wide one. Then we examined how our model's performance scaled as we increased the complexity of the training dataset. Table 2 and in Figure 3 summarize how model performance consistently degrades as the input functions become more complex. Next, we studied how large (number of trainable parameters) operator learning models need to be to achieve good generalization on out-of-distribution inputs. We found that our model makes worse predictions for more complex out-of-distribution inputs and better predictions for smoother, simpler inputs, as shown in Figure 4. Furthermore, we found that model size needs to be increased substantially in order to achieve good generalization; Figure 5 demonstrates the trend of gradually increasing performance in terms of model size. We also observed that our model's prediction error is relatively independent of the amount of time between the initial and final conditions, as shown in Table 3. Finally, we showed that our operator learning model can predict wave propagation equally well in anisotropic materials as in isotropic materials; example inputs and outputs for anisotropic materials with different types of symmetry are shown in Figure 6.

Based on the results described above, we realized that the challenges we observed are all related to the hyperbolic (wave-like) nature of the problem. We then developed a mathematical explanation for these phenomena by establishing that hyperbolic problems conserve the variance (i.e., complexity) of the input distribution over time, leading to the observed challenges for operator learning. By contrast, we also showed that in parabolic (diffusive) systems variance decays over time, making operator learning more effective and efficient. Figure 7 shows the qualitative difference in complexity between hyperbolic and parabolic problems.

Since these insights are important enough to expose for the broader operator learning community, we developed examples to explore the challenges posed by hyperbolic problems in general. First, we developed examples for the 1D wave and heat equations. Figures 8 and 9 demonstrate how operator learning is more effective on parabolic problems, with accuracy improving in relation to the decay of the variance, supporting the link between variance and operator learning performance. Second, we developed examples for the 2D advection-diffusion equation, in which we vary the strength of the diffusion term, gradually transitioning from hyperbolic to parabolic, as shown in Figure 10, which will allow us to show how operator learning performs better for more parabolic problems. In addition, we implemented numerical solvers to generate datasets for a nonlinear wave problem, shown in Figure 11, as well as for classical elasticity, shown in Figure 12, to bolster our evidence for the challenges of hyperbolic problems for operator learning. We have also collected these results into a nearly complete manuscript draft.

Accomplished goals

As stated in our proposal, there were three goals for this part:

1. Extension of model to vector-valued case

2. Generation of benchmark datasets
3. Evaluation of operator learning effectiveness

All three of these goals was achieved, as described in the outcomes listed above, as well as additional accomplishments based on the insights gained while pursuing the stated goals.

Contributions

- Valuable and varied benchmark datasets for elastic wave propagation
- Introduction of operator learning as a form of surrogate modeling in peridynamics
- Mathematical theory as well as numerous numerical experiments to illustrate how hyperbolic problems lead to significantly less efficient operator learning performance compared to parabolic problems

Crack propagation

In this part we applied our operator learning model to fracture problems. Here, the training data consists of pairs of input and output functions, where the input describes the loading forces on a sample, and the output is the damage field that results from an initial notch propagating into a larger crack caused by the force on the sample.

Outcomes and conclusions

We developed three benchmark datasets for crack propagation using PDMATLAB2D based on predicting crack path from loading conditions. Figure 13 shows examples from these datasets. Figure 14 shows a successful operator learning prediction from our initial investigations of operator learning using our benchmark datasets. In our investigations, we observed that the sensitivity of the particular brittle fracture scenario significantly limits the effectiveness of operator learning. In order to better understand the nature of the sensitivity, so that we might potentially address it, we demonstrated that the crack path is highly sensitive both to the numerical discretization, as seen in Figure 15, as well as to the loading force, as seen in Figure 16. Successful operator learning, however, depends on stability with respect to both the input (loading force, in this case) and the numerical discretization.

In addition to our studies of 2D crack propagation, we also developed tools for polycrystal fracture simulation in CabanaPD. A polycrystal is a solid made of many smaller crystal grains; our primary result in this part of the project was the development of two mechanisms for generating polycrystal grain structures. First, we developed a custom Voronoi grain generation algorithm--an example grain structure and corresponding fracture simulation are shown in Figure 17. Second, we coupled CabanaPD to the code ExaCA, which uses a solidification simulation to generate realistic grain structures--an example structure and fracture simulation are shown in Figure 18.

Accomplished goals

As stated in our proposal, there were three goals for this part:

1. Development of benchmark crack propagation datasets
2. Investigation of potential challenges for operator learning and development of techniques for handling discontinuities
3. Development of simulation tools in CabanaPD for eventual extension to more complex, 3D problems

While we were able to partly achieve goals 1. and 2. by generating initial crack propagation datasets and evaluating operator learning methods on these datasets, challenges with crack sensitivity prevented us from finalizing our datasets and exploring potential approaches for dealing with discontinuities in operator learning. We completed our goal of developing tools in CabanaPD for more complex data generation.

Contributions

- Increased understanding of the limitations of operator learning in the context of fracture mechanics, and direction for future work
- Development of polycrystal fracture simulation tools that will be useful for both my future work and current and future projects at ORNL

Challenges

Challenges for wave propagation

- Sharp cost-accuracy trade-off and model parameter inefficiency

Challenges for fracture mechanics

- Discovery of issues with sensitivity meant that intended development of operator learning methods for fracture was not completed, and crack propagation benchmark datasets are not finalized

Future Work

Future work for wave propagation

- Complete experiments about hyperbolic problems and finish corresponding manuscript
- Develop second paper introducing operator learning to peridynamics through our wave propagation examples and further examples

Future work for crack propagation

- Modify current benchmarks to more stable setting to enable effective operator learning
- Explore issues and develop operator learning techniques for problems involving discontinuities

CUI

- Apply operator learning to polycrystal setting using CabanaPD to generate data

Uploading Additional Materials

Would you like to upload any additional materials?

Yes

Additional Materials: main.pdf

Use of DOE Laboratory Resources

DOE Laboratory Facilities and Research Capabilities:

The main use of DOE lab facilities was for office space and utilities, such as conference rooms. Being able to work in-person with lab staff, especially my project mentor, has been essential to the success of the project.

Additionally, I was able to use the Atlasmath high-performance cluster, whose high-memory graphics processing units enabled some of my computational experiments.

Reporting Changes or Issues

Were there any changes or issues in accomplishing the research goals and objectives?

No

PRODUCTS

Products as a Direct Result of the SCGSR Project

Do you have any completed products?

No

Planned Products and/or Products Under Development

Are you developing or planning to develop any type(s) of the above mentioned research products as a direct result of the SCGSR research project?

Yes

Publication Type: Archival Publication - Peer Reviewed Scientific or Technical Journal Article

Working Article Title: Why Hyperbolic Partial Differential Equations are Challenging for Operator Learning

Author(s): Jacob Hauck, Pablo Seleson, Yanzhi Zhang

Tentative Journal Name: Journal of Machine Learning Research

Publication Volume:

Publication Issue Number:

Publication Page Numbers:

Publication Year:

Digital Object Identifier (DOI) or URL:

Publication Type: Archival Publication - Peer Reviewed Scientific or Technical Journal Article

Working Article Title: Accelerating Nonlocal Mechanics Simulations with Operator Learning Surrogates

Author(s): Jacob Hauck, Pablo Seleson, Yanzhi Zhang

Tentative Journal Name: TBD

Publication Volume:

Publication Issue Number:

Publication Page Numbers:

Publication Year:

Digital Object Identifier (DOI) or URL:

IMPACTS

Contribution to Thesis / Dissertation

CUI

Research Project Progress:	This project has made three significant contributions to my thesis. First, it provided an important application area to demonstrate the effectiveness and applicability of the methods developed in my thesis. Second, the study of fracture mechanics in this project provided motivation to develop and test new approaches to operator learning for problems characterized by discontinuous functions. Third, the work on wave propagation in this project transformed into a larger study on operator learning for wave-like problems. Each of these will be an major component of my thesis.
-----------------------------------	---

Graduate Training and Education

Knowledge and Skills:	Following are skills and knowledge that I either gained or significantly developed through my project. • Software: CabanaPD, ExaCA, Kokkos, Paraview, Slurm, Linux • Technical knowledge: solid mechanics, peridynamics, high-performance computing • Research skills: project management, leading group discussions, proposal writing, paper writing, designing and giving oral presentations • Other: increased self-confidence
------------------------------	---

Development of the Principle Discipline

Impacts of Principle Discipline(s):	This project will contribute to two fields: operator learning, which studies machine learning methods to accelerate scientific simulations; and nonlocal mechanics, which is important for modeling fracture in materials. Our results on operator learning for wave phenomena will improve understanding of challenges for studying wave problems, opening the door to improve operator learning methods. Our application of operator learning to nonlocal mechanics introduces new methods to accelerate simulations, and the tools we developed for fracture simulations will be useful in current and future projects.
--	---

OTHER ACTIVITIES

Participation in Other Activities at the Host DOE Laboratory During the SCGSR Project Period

Participation in research Seminars/Workshops within the division you are currently:	Occasionally
Participation in Research Seminars/Workshops in other divisions:	Occasionally
Participation in Training Seminars/Workshops on Professional Development:	Occasionally
Seminars/Workshops description:	Some of the events that stood out to me were the talks by former SCGSR participants discussing their career paths. I was able to attend several of these, learning about how the participants' experiences in the SCGSR program as well as in their PhD in general affected their eventual careers. In one talk I learned about a participant's transition to work in industry. In that talk I learned how they used their unique experience working in a state-of-the-art research during the SCGSR program gave them a competitive edge when entering industry and how they used this as a springboard to gain access to more unique work experiences. Another memorable talk was by a participant who co-founded a startup company based on their PhD research. To me the most significant message was the importance of practicing presenting your research to a broader audience, which holds for any career path, but especially for forming a startup that will live or die by your ability to persuade investors.

ABSTRACT

General Audience Abstract

General Audience Abstract:	I performed my SCGSR project at Oak Ridge National Lab (ORNL), working with Dr. Pablo Seleson. During my project I saw firsthand how the high-performance computing facilities that ORNL is known for are being used to run complex simulations of how materials deform and break. Simulations like these allow researchers to better understand existing materials and to design better materials that can improve infrastructure and enable new technologies. Complex simulations, however, cost time, money and energy, limiting how quickly we can study and develop new materials. In my PhD work I have developed new methods for operator learning, a machine learning technique that can significantly accelerate scientific simulations--including material simulations--by a factor of several hundred times. During my project at ORNL I worked on applying my operator learning methods to accelerate material simulations. In my time at ORNL I wrote code, performed mathematical analyses, developed simulations, and trained machine learning models. I was able to successfully apply operator learning to predict how deformation waves propagate in a simple material, which also led to significant insights about the challenges for operator learning for studying
-----------------------------------	---

wave-like phenomena. Furthermore, I began initial investigations of operator learning for predicting cracks. I also developed tools for simulating fracture in complex materials called polycrystals, which are made of many individual crystal grains and are highly relevant in real industrial settings, for example, in additive manufacturing. These tools will enable my future study of operator learning.

Overall, my experience at ORNL was invaluable, contributing significantly to my PhD research, giving me experience developing software for high-performance computing, increasing my knowledge in material science, as well as growing my professional network and helping me to develop professional skills like project management and communication.

COMMENTS

Reflection or Comments About Your Overall SCGSR Experience at the Host DOE Laboratory

Would you like to provide reflection or comments about your overall SCGSR experience at the Host DOE Laboratory?

No

SUBMIT DATE

Submit Date:

SCGSR Final Report Additional Materials

Jacob Hauck

August 30, 2026

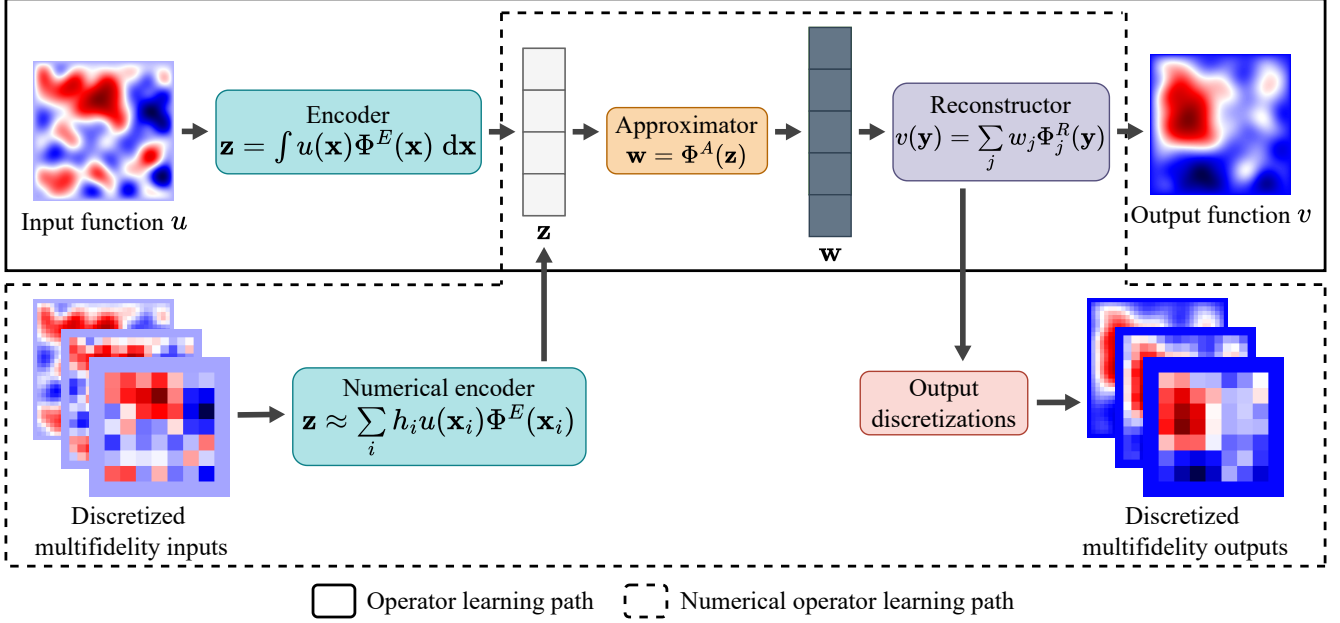


Figure 1: The architecture of our operator learning method, and how operator learning enables discretization independence through numerical operator learning. Note the three-module structure: (linear) encoder projects the input function onto a finite-dimensional space parameterized by neural network basis functions (Φ^E), approximator approximates the operator in the latent space, and (linear) reconstructor builds the output function by taking a linear combination of neural network basis functions (using a different neural network Φ^R) using the latent output variables from the approximator as weights. Vector-valued extension is described in (1).

$$\begin{cases} E(\theta_E; u) = (\langle u, \Phi_1^E \rangle, \dots, \langle u, \Phi_p^E \rangle) \in \mathbb{R}^p, & u \in \mathcal{U} = L^2(\Omega_{\mathcal{U}}; \mathbb{R}^{c_{\mathcal{U}}}), \\ A(\theta_A; \mathbf{z}) = \Phi^A(\mathbf{z}) \in \mathbb{R}^q, & \mathbf{z} \in \mathbb{R}^p, \\ R(\theta_R; \mathbf{w}) = \sum_{j=1}^q w_j \Phi_j^R \in \mathcal{V} = L^2(\Omega_{\mathcal{V}}; \mathbb{R}^{c_{\mathcal{V}}}), & \mathbf{w} \in \mathbb{R}^q, \end{cases}$$

where

$$\Phi^E(\mathbf{x}) = \begin{bmatrix} \varphi_1^E(\mathbf{x}) \\ \varphi_2^E(\mathbf{x}) \\ \vdots \\ \varphi_p^E(\mathbf{x}) \end{bmatrix}, \quad \mathbf{x} \in \Omega_{\mathcal{U}}, \quad \Phi^R(\mathbf{y}) = \begin{bmatrix} \varphi_1^R(\mathbf{y}) \\ \varphi_2^R(\mathbf{y}) \\ \vdots \\ \varphi_q^R(\mathbf{y}) \end{bmatrix}, \quad \mathbf{y} \in \Omega_{\mathcal{V}},$$

$$\varphi_i^E : \Omega_{\mathcal{U}} \rightarrow \mathbb{R}^{c_{\mathcal{U}}}, \quad \varphi_i^R : \Omega_{\mathcal{V}} \rightarrow \mathbb{R}^{c_{\mathcal{V}}}.$$

(1)

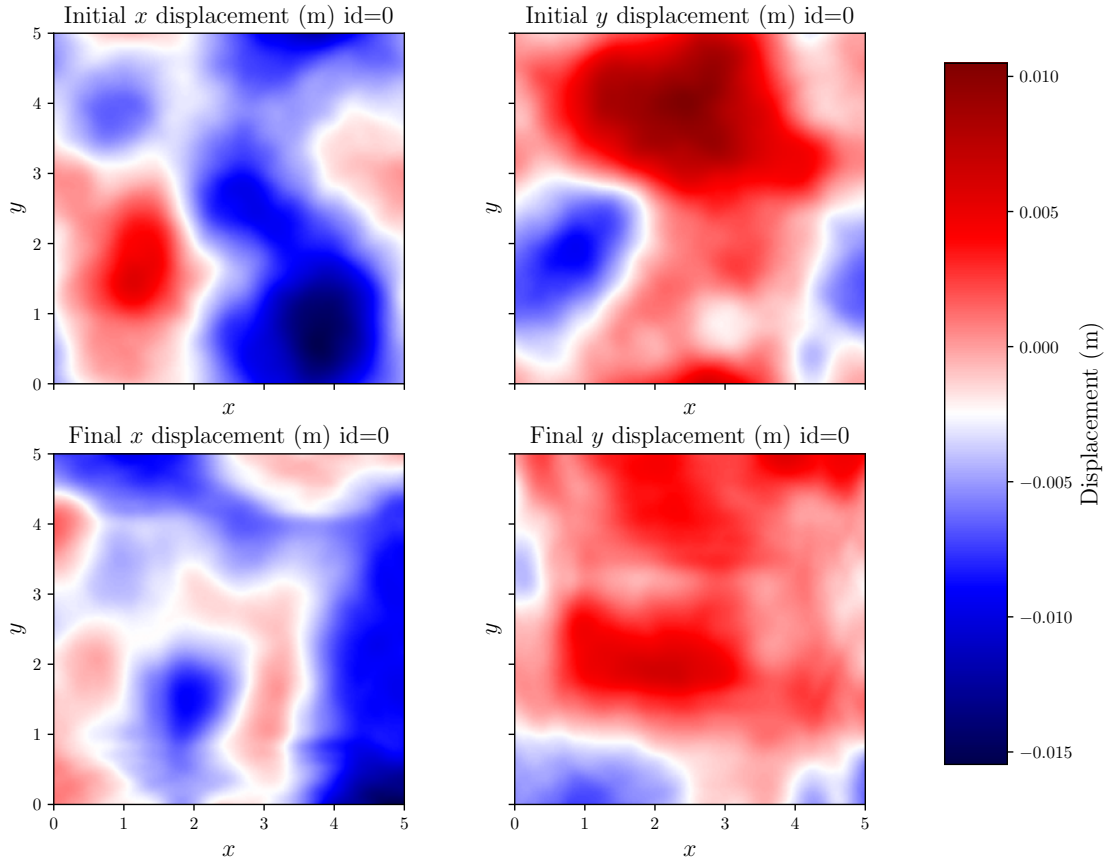


Figure 2: Example input–output pair from one of the wave propagation benchmark datasets, prepared using PDMATLAB2D. The input function is the top row, which shows the initial x and y displacement fields. The output function is the bottom row, which shows the final x and y displacement fields after a time $T = 1.8$ seconds has elapsed in the simulation. The “id=0” refers to this sample’s ID in the dataset.

Approximator	Error on training data	Error on testing data
Base (3 layers, 128 neurons/layer, ReLU)	15.2	18.1
Wide (3 layers, 384 neurons/layer, ReLU)	12.4	20.7
Deep (8 layers, 128 neurons/layer, SELU)	11.9	16.9

Table 1: Error (measured in % relative L^2) for different approximator network settings. Using a wide approximator leads to overfitting (note gap between train and test error in the second row), but using a deep approximator with self-normalizing activation leads to a slight performance boost (compare error in first and third rows).

γ	Error on training data	Error on testing data
2.5	14.3	18.6
3.1	7.6	10.7
4.0	4.6	7.1
5.0	4.0	5.7

Table 2: Error (measured in % relative L^2) for different input distribution complexities, as measured by the γ decay rate of the covariance eigenvalues; smaller γ means more complex. We observe that the more complex data is more difficult for operator learning models to fit, as measured both on the training and testing data.

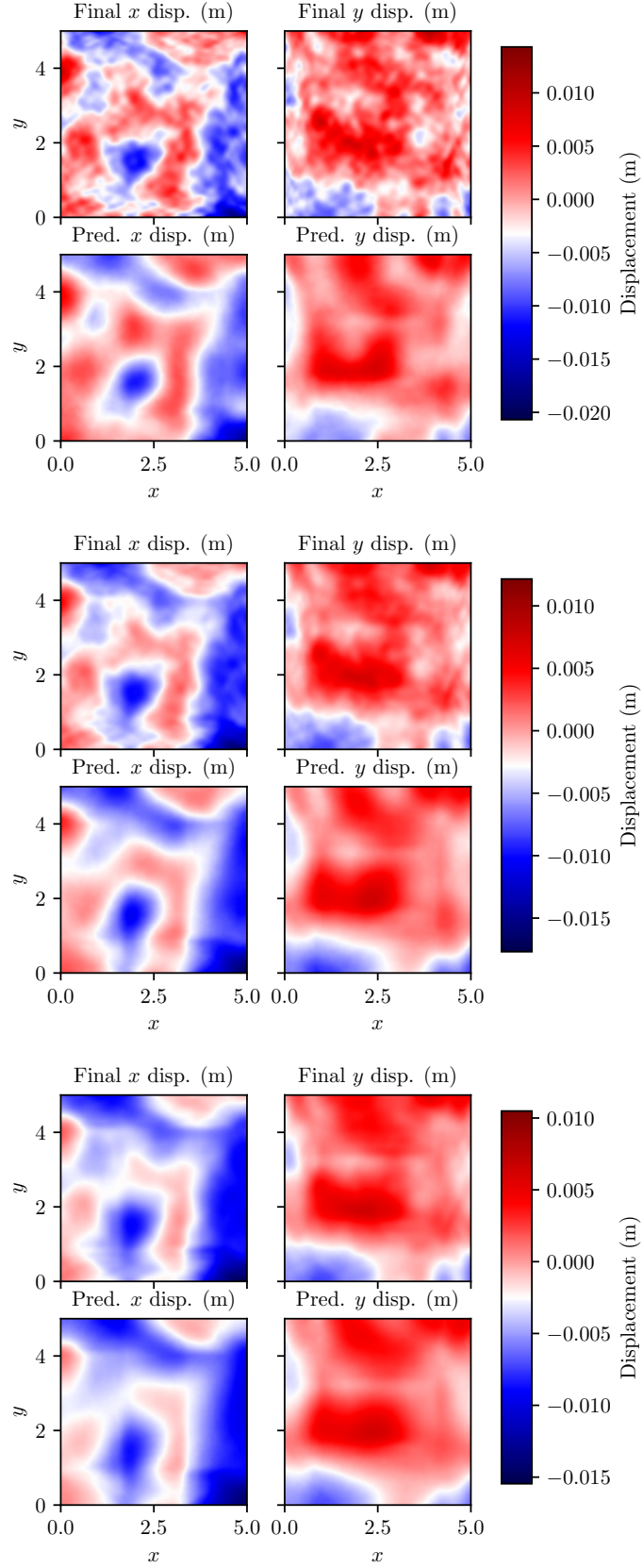
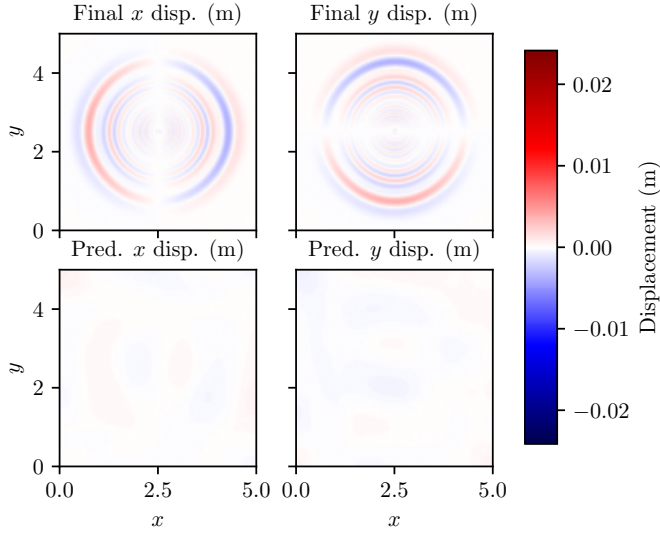
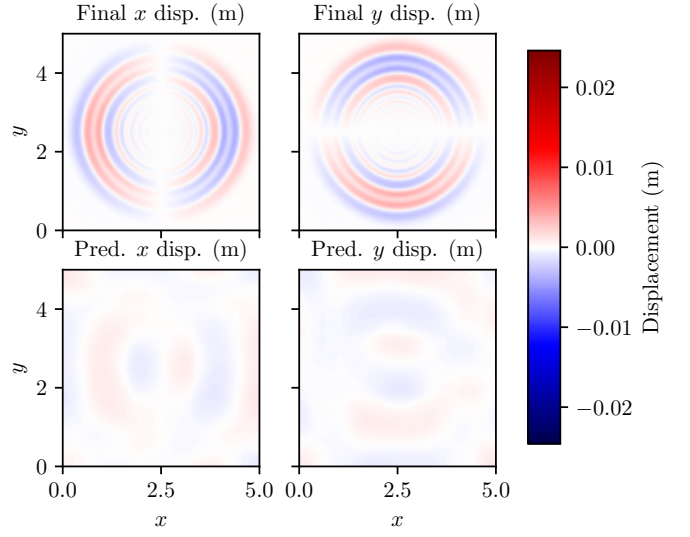


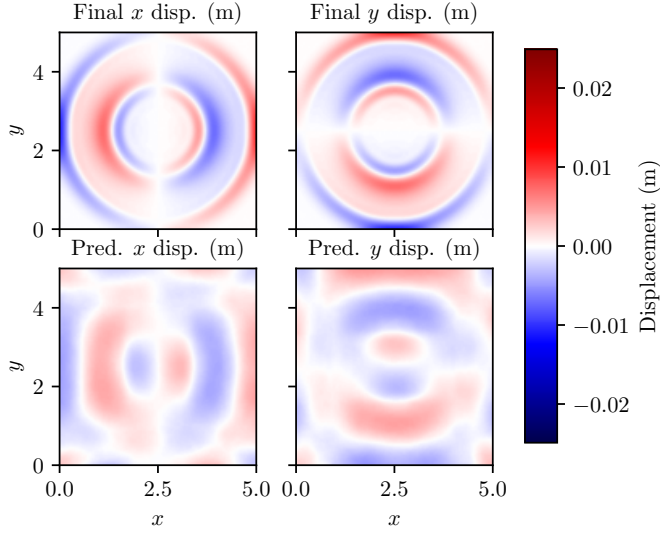
Figure 3: Example final displacement fields (top rows, x and y components alternating) and corresponding model predictions (bottom rows) for different input Gaussian process covariance eigenvalue decay rates. Decay rate increases from left to right. Note the corresponding decrease in strong high-frequency components from left to right. Also note how the model prediction only capture low-frequency (smooth) components, making its predictions more accurate when the covariance eigenvalues decay quickly.



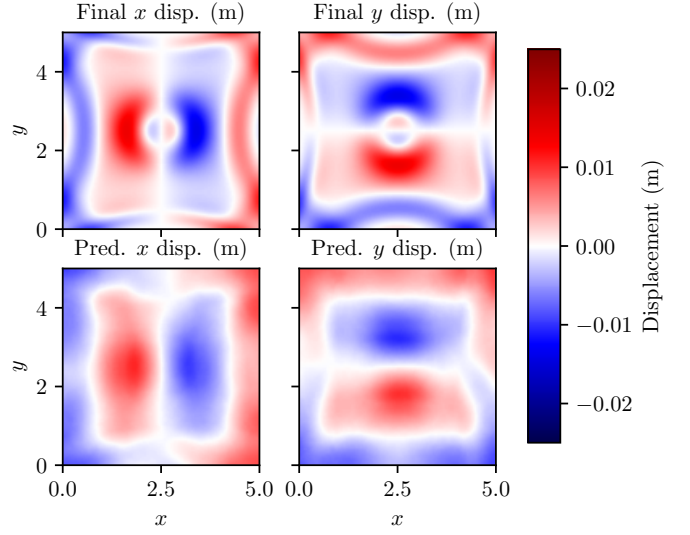
(a) Initial displacement scale = 0.03



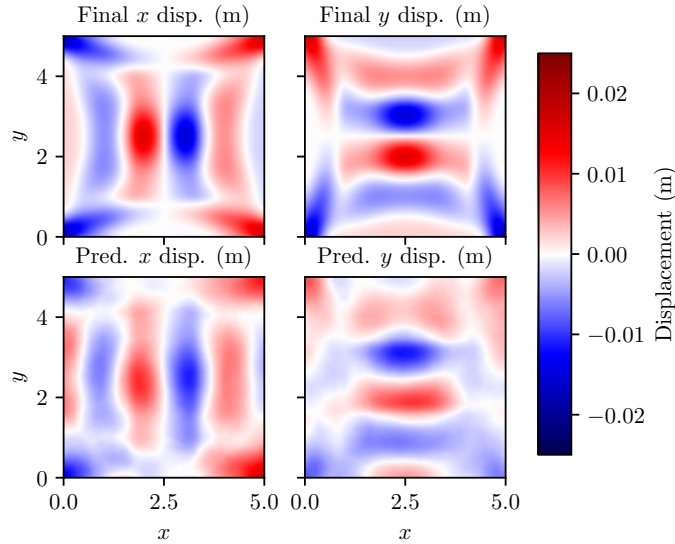
(b) Initial displacement scale = 0.05



(c) Initial displacement scale = 0.10



(d) Initial displacement scale = 0.20



(e) Initial displacement scale = 0.30

Figure 4: Model predictions for out-of-distribution test initial displacements. The top row in each block shows simulation final displacement field x and y components, and the bottom rows show the corresponding model predictions. Input initial displacement fields are Gaussian bumps of varying sizes, specified in the captions of each subfigure. The model makes more accurate predictions for larger bumps (which have less high-frequency information and result in less dispersive effects).

T	Error on training data	Error on testing data
1.8	15.4	18.2
1.4	16.5	21.0
1.0	16.0	19.3
0.6	14.1	17.1
0.2	14.4	19.2

Table 3: Independence of model error (measured in % relative L^2) on simulation duration T . When $T = 0.2$, the operator is nearly the identity, suggesting that the reconstructor module in our model is insufficient. As illustrated in Figure 5, the error of our model can be reduced by increasing the number of basis functions in the encoder and reconstructor.

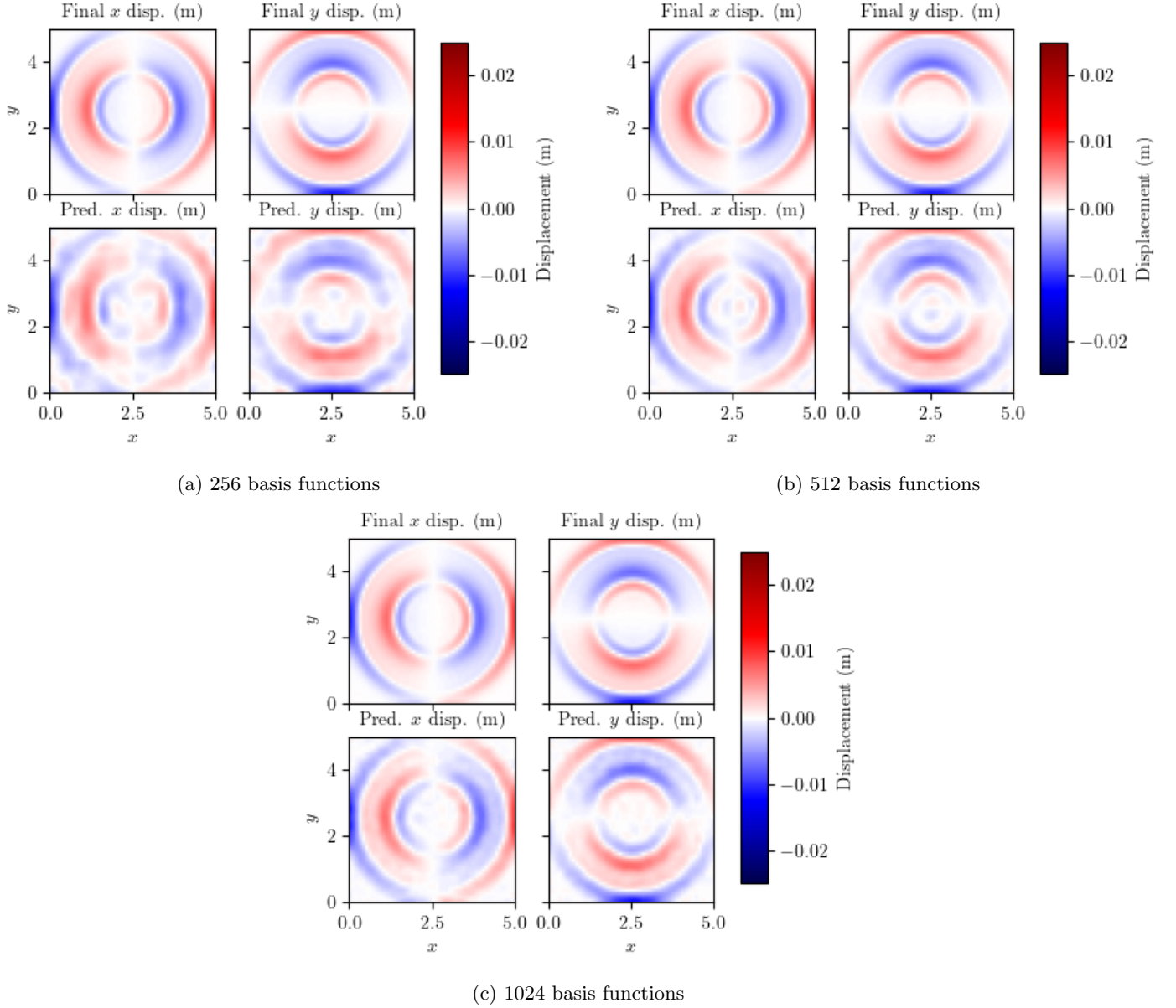


Figure 5: Shows out-of-distribution predictions (bottom rows) for different model sizes compared to true final states (top rows), as measured by the number of basis functions in the encoder/reconstructor modules (p and q in (1)). We need to increase the number of basis functions substantially in order to achieve good accuracy, which is a symptom of the challenges of hyperbolic problems (such as elasticity) for operator learning.

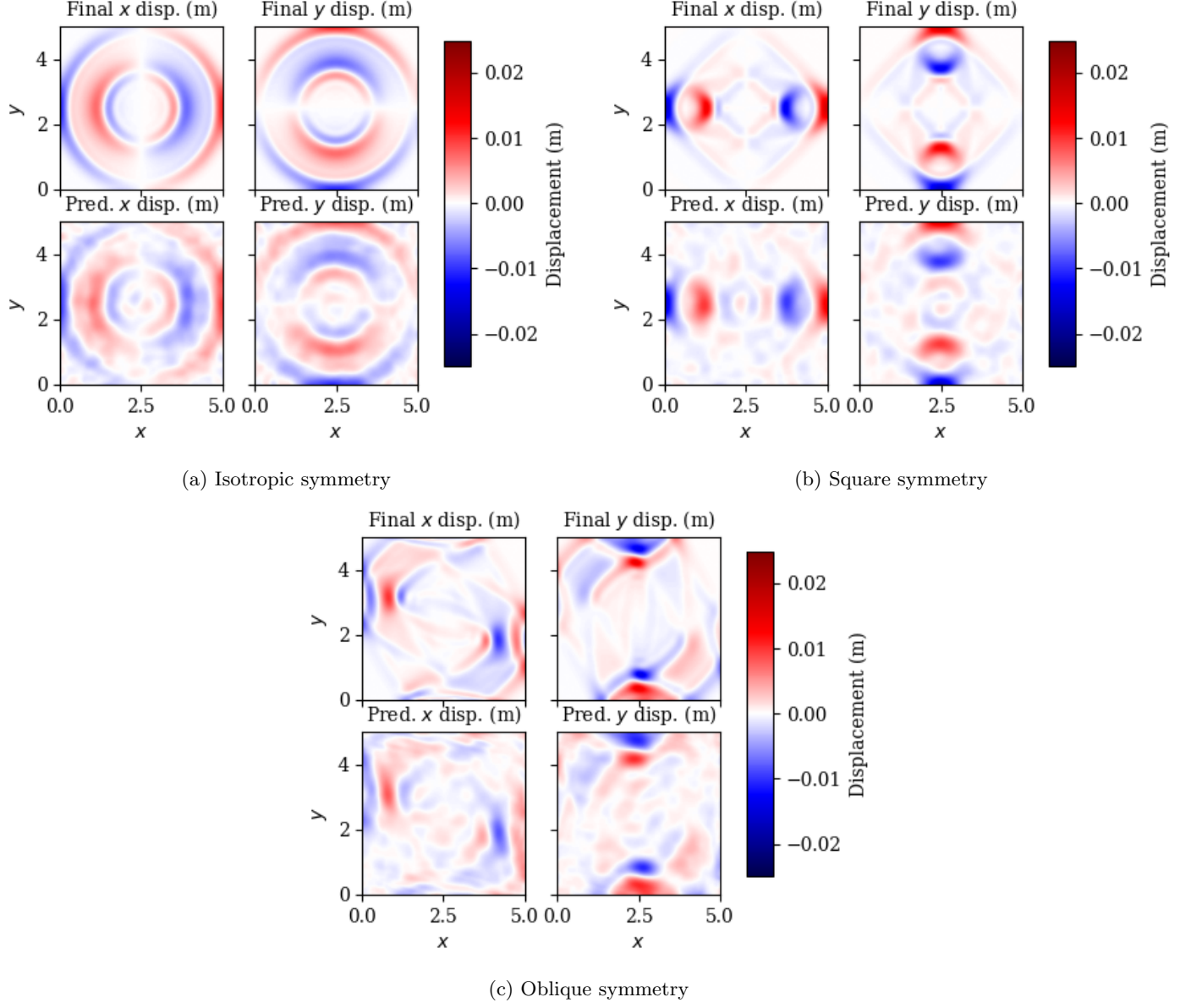


Figure 6: Final conditions under different material symmetry models resulting from the same initial condition, along with operator learning predictions from models trained on datasets with corresponding symmetry. Top rows: true final states. Bottom rows: operator learning predictions. We observe similar accuracy across all symmetry types.

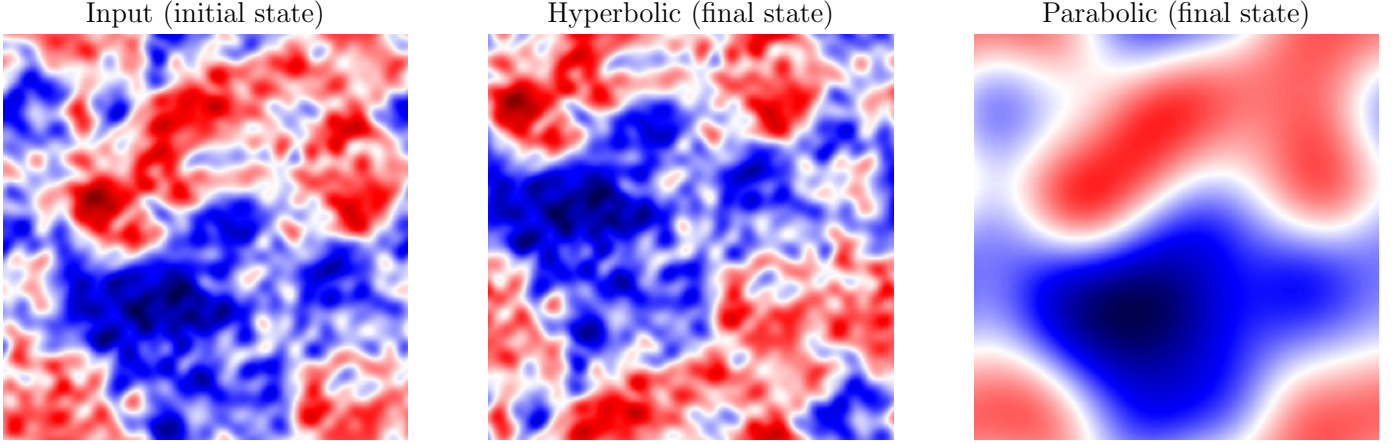


Figure 7: Comparison of propagation operators for hyperbolic and parabolic systems. The final state in the hyperbolic problem retains complex (high-frequency) features in the initial state whereas the final state in the parabolic problem is significantly simpler, with all the fine details smoothed out.

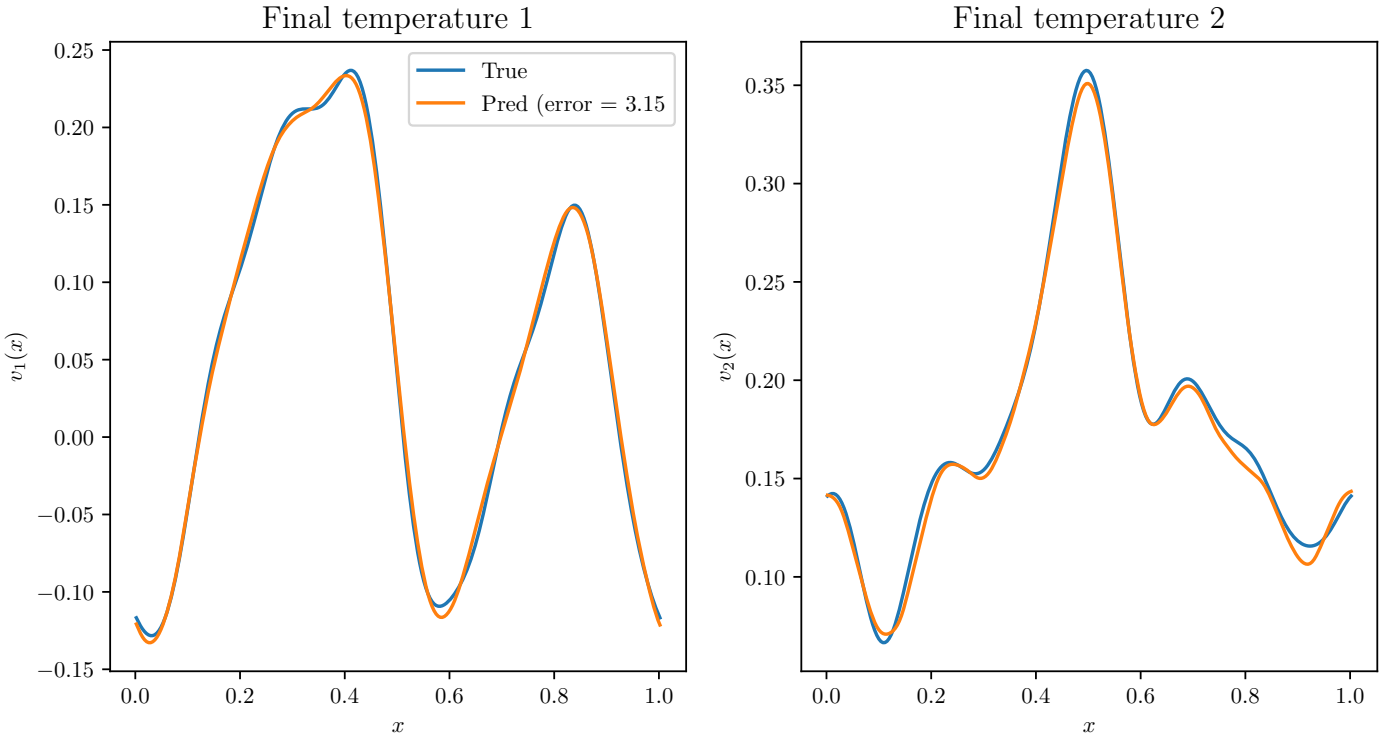


Figure 8: Example final state for two instances of the 1D heat equation. We can see that the final states are smoother than in the wave case in Figure 9, and the operator learning model makes quite accurate predictions (3.15% relative L^2 error).

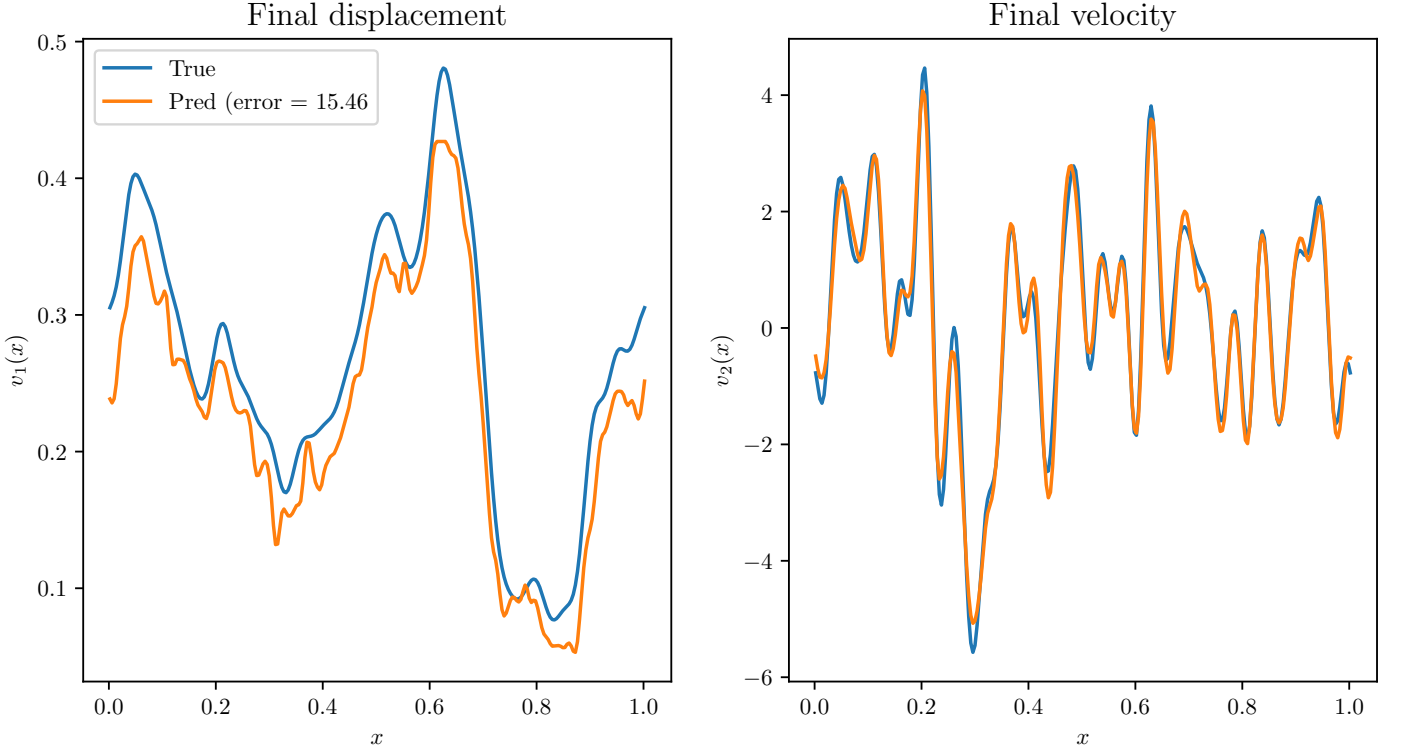


Figure 9: Example final for two instances of the 1D wave equation, using the same initial state as in the heat equation in Figure 8. We can see that the final states here are more oscillatory than in the heat equation in Figure 8, and the operator learning model makes less accurate predictions (15.46% relative L^2 error).

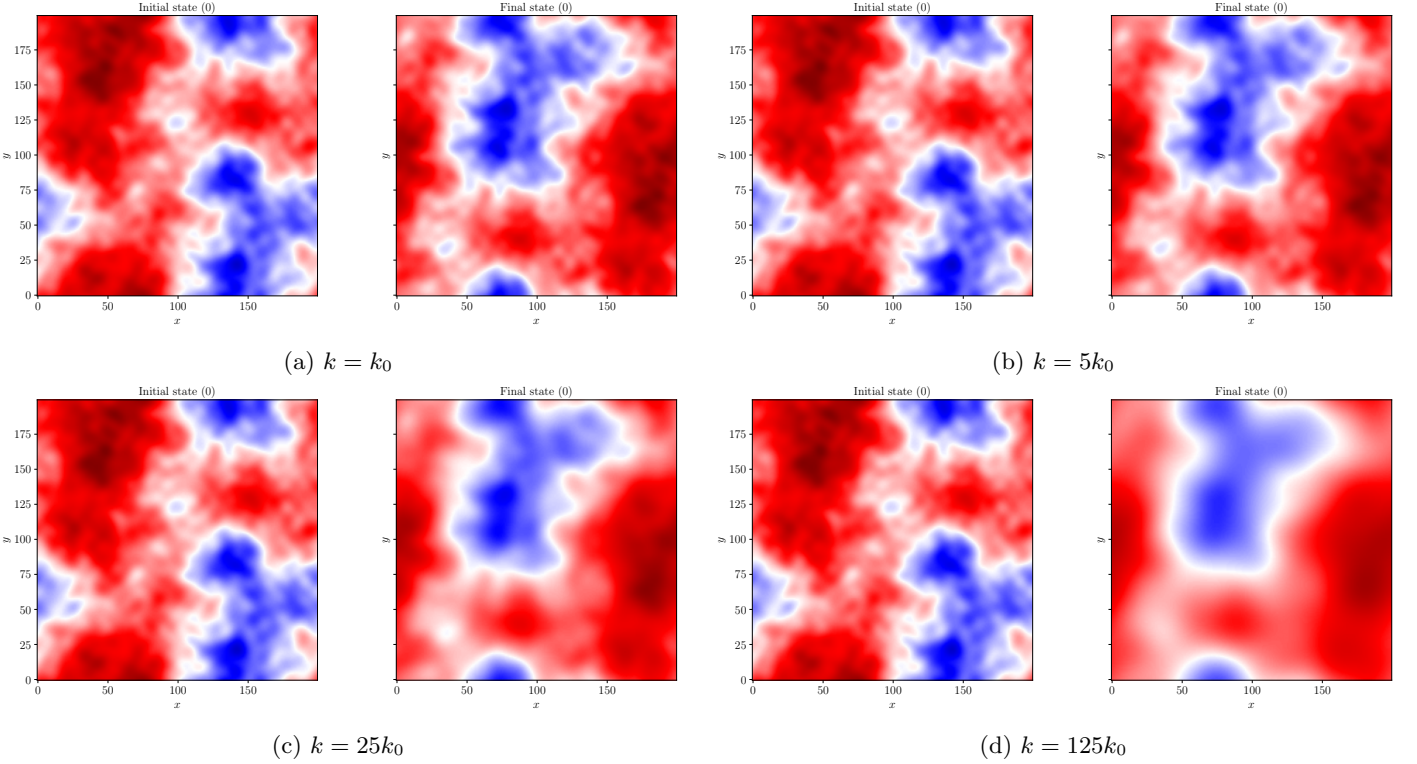


Figure 10: Sample input and output functions for the 2D advection–diffusion equation with different values of the diffusion constant k . Since the same seed was used to generate each dataset, the initial state (left function in (a–d)) is the same in all cases, but the final state (right function in (a–d)) becomes smoother as k increases because the physics become more diffusion-dominated. We expect to use these datasets to show that operator learning consequently has higher performance for higher values of k , that is, when the hyperbolic behavior of the advection term is more dominated by the parabolic behavior of the diffusion term.

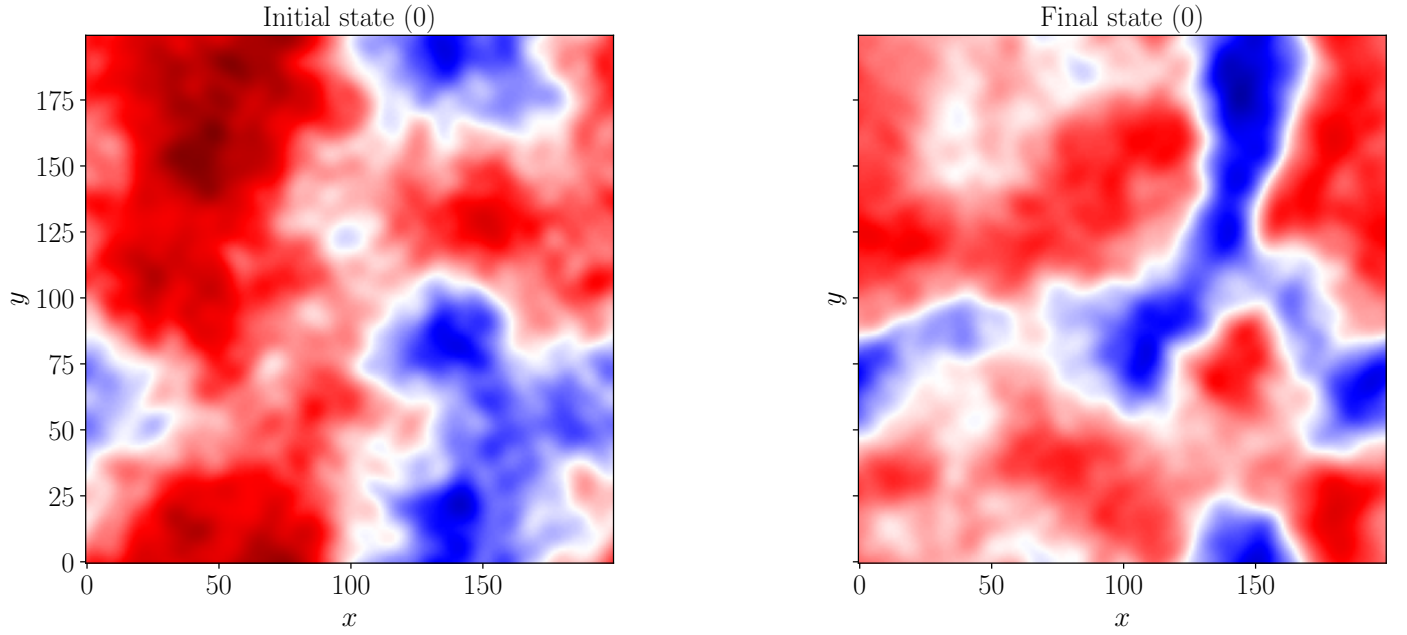


Figure 11: Sample initial and final states from the nonlinear wave equation dataset. For planned operator learning experiments, the input is the left function and the output is the right function.

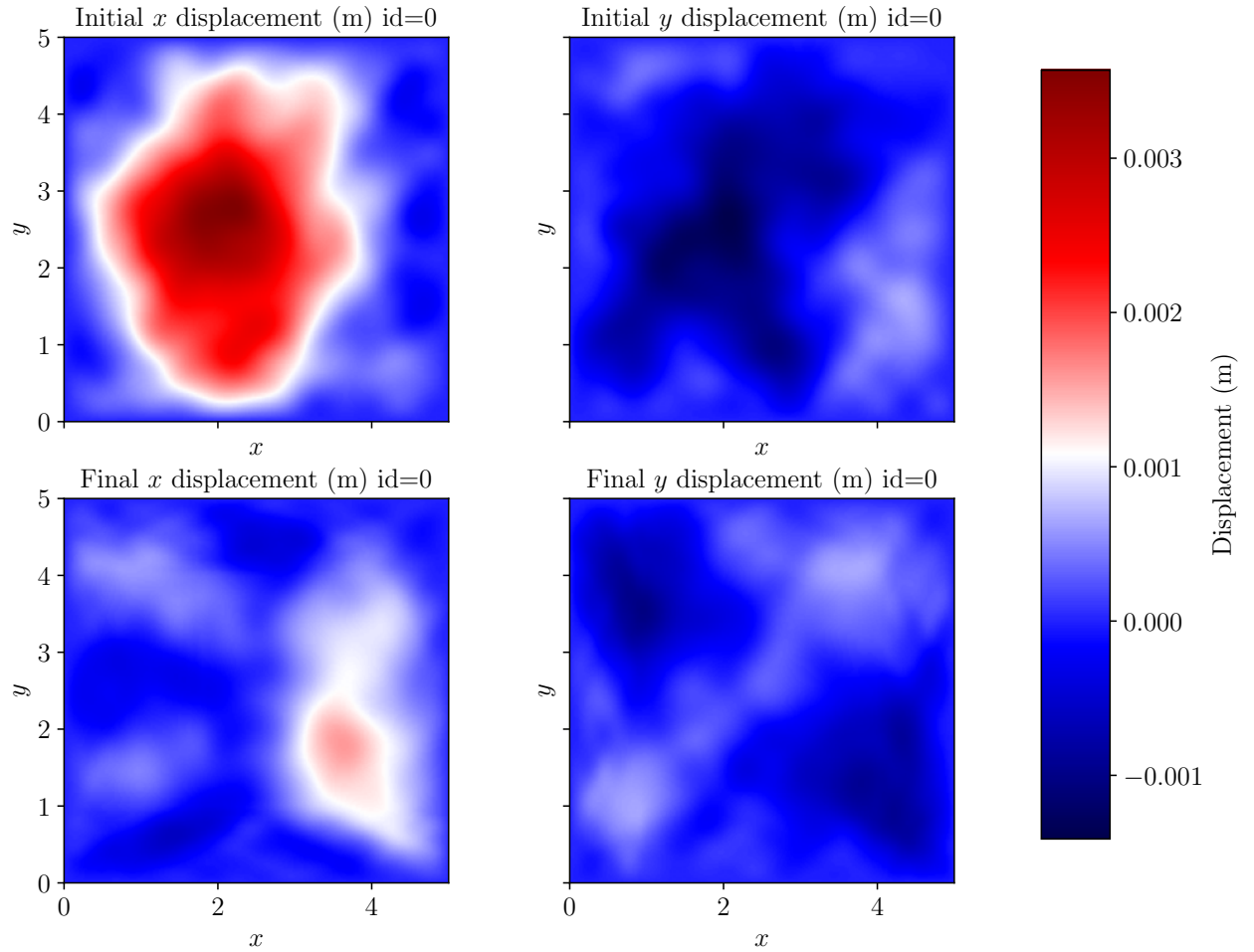


Figure 12: Sample initial and final states from the classical elasticity dataset. For planned operator learning experiments, the input is the top row and the output is the bottom two row; the left column is the x displacement, and the right column is the y displacement.

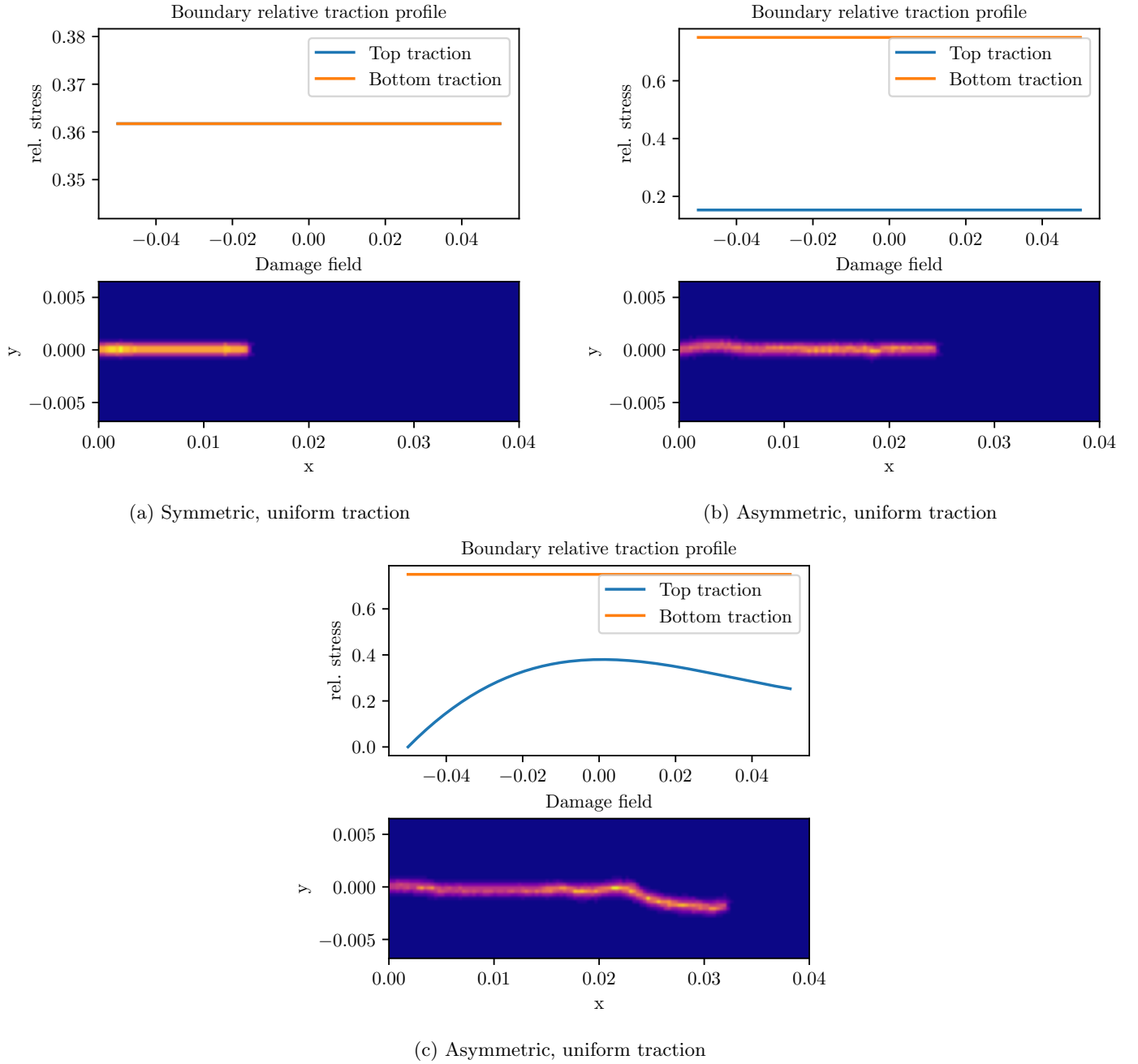


Figure 13: Samples from generated crack propagation datasets. The input function is the boundary relative traction profile (top row, units are relative to a fixed, maximum traction), and the output function is the final damage field (bottom row, cropped to the region around where the crack grew). Note that dark blue means no damage, and orange/white means high damage. The goal for operator learning on these datasets is to predict the final crack path.

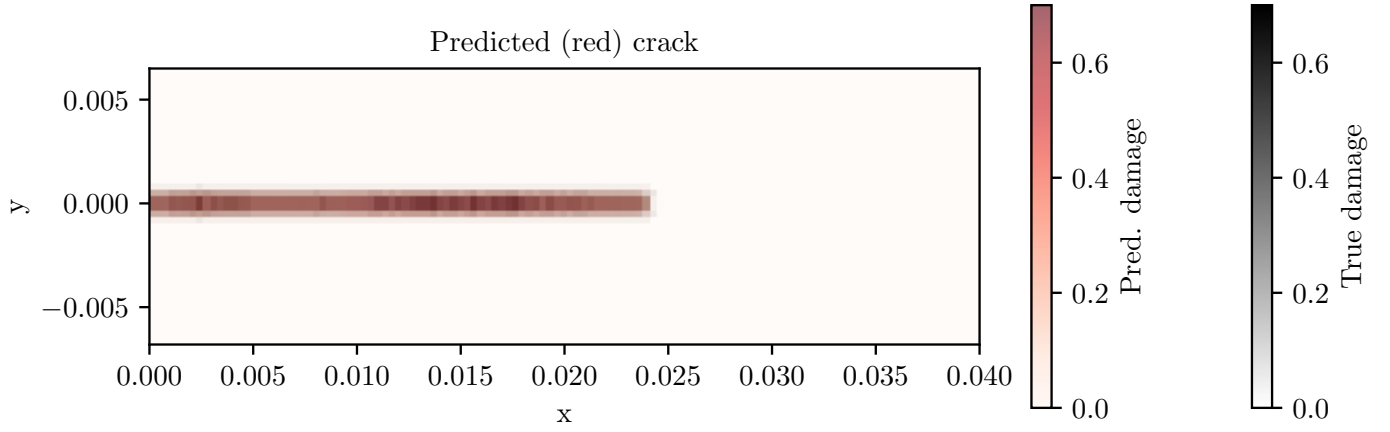


Figure 14: An example crack final damage field (black) and model prediction superimposed (red) in the case of symmetric traction on the top and bottom of the plate. This image is cropped to show only the part of the crack that grew. Note that the damage field refers to the fraction of broken bonds (between 0 and 1) per particle. The model prediction agrees closely with the true crack path.

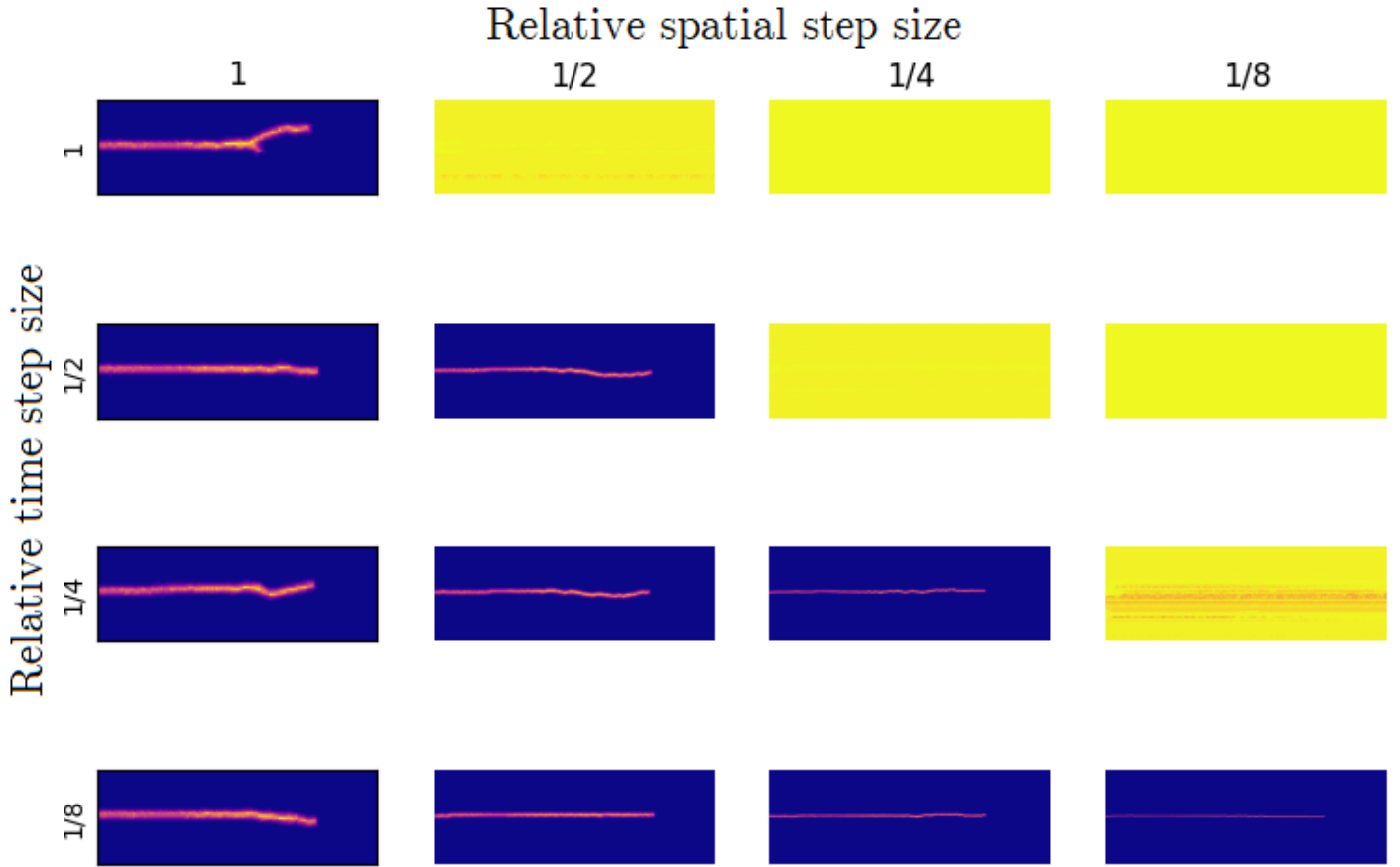


Figure 15: Study of numerical convergence of crack propagation with asymmetric, constant loading with respect to discretization parameters. Each column corresponds to a fixed spatial step size Δx and each row corresponds to a fixed temporal step size Δt in the numerical simulation. Yellow indicates numerically unstable cases. We expect to see convergence to a common solution as Δx and Δt get smaller, but each combination produces inconsistent results.

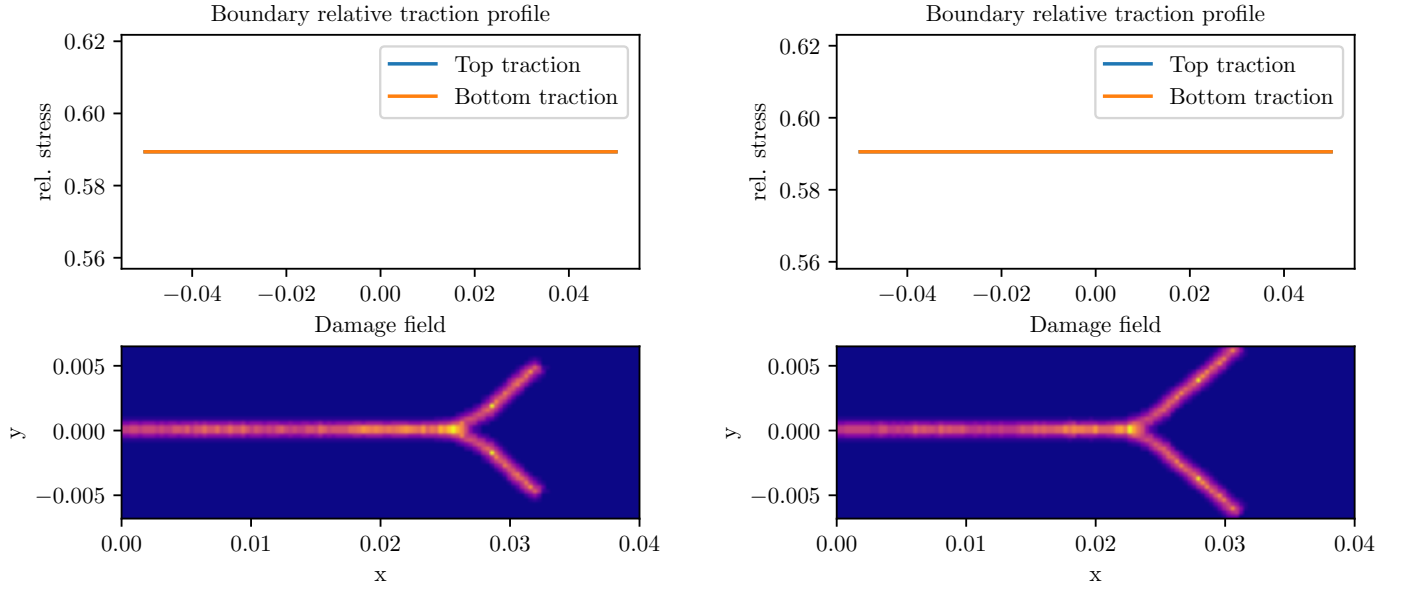
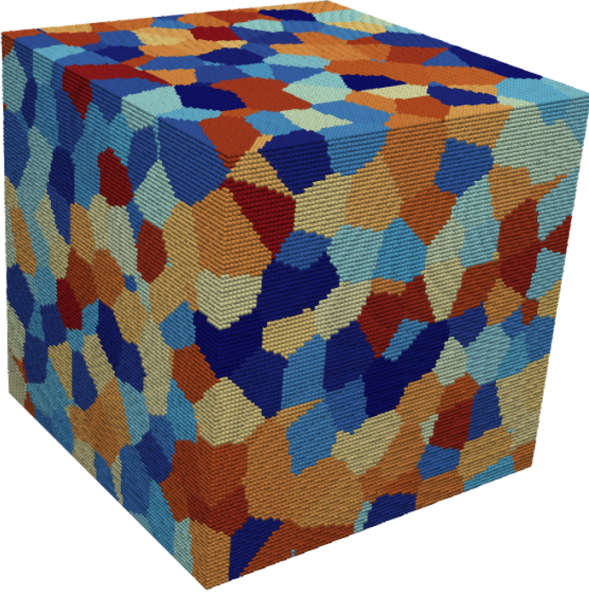


Figure 16: Comparison of simulated crack paths with symmetric, uniform relative traction that differs in strength by less than 1%. The simulation produces very different results despite the nearly identical parameters. This makes prediction with operator learning nearly impossible, so finding a stable configuration is a top priority for future work following my internship.

Generated grain structure
(uniform size, isotropic shape)



Simulated evolution of damage

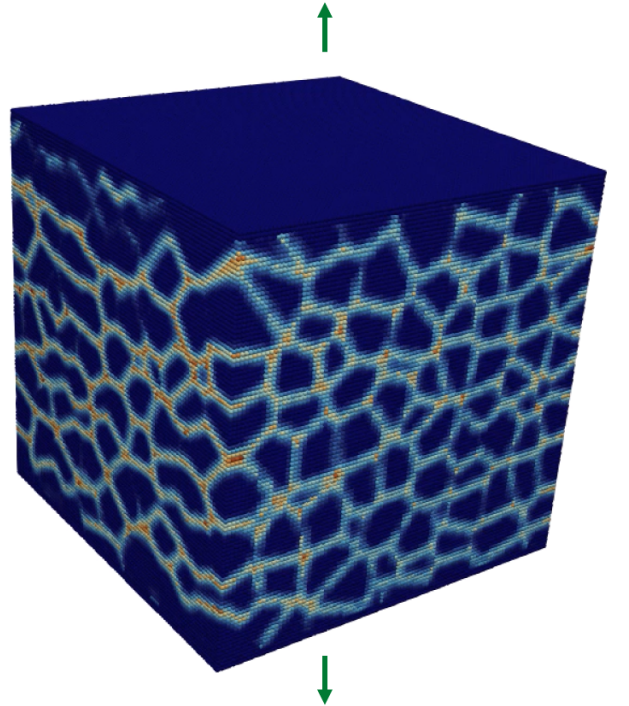
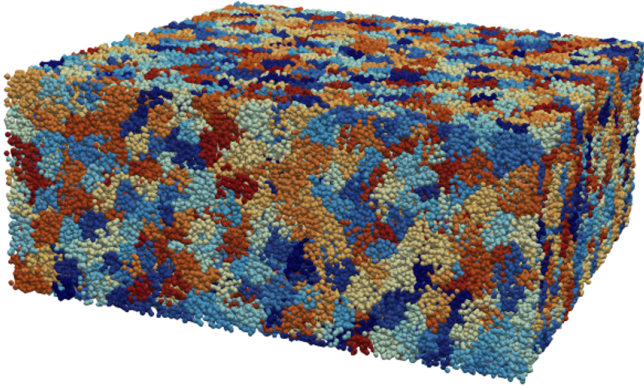


Figure 17: Left: example grain structure from my custom Voronoi tessellation algorithm that is integrated into CabanaPD. Right: final damage field (orange and yellow indicate high damage, blue indicates no damage) simulated in CabanaPD as a result of vertical tensile loading. Note how the damage pattern reflects the grain boundaries.

Grain structure from ExaCA



Evolution of damage (from CabanaPD)

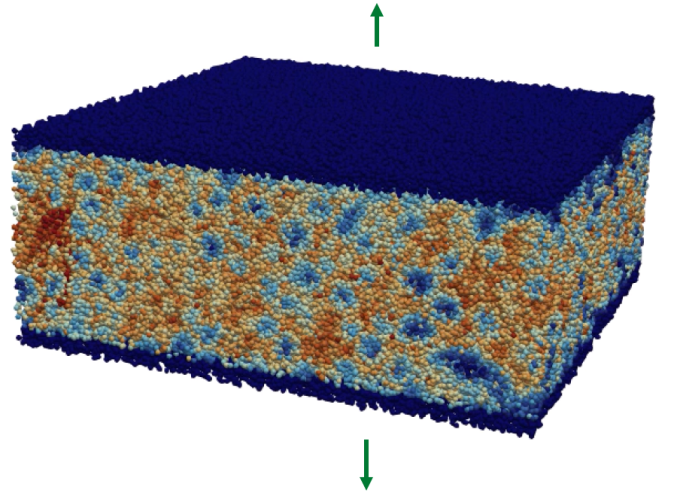


Figure 18: Left: example grain structure generated using ExaCA. Right: final damage field (orange and yellow indicate high damage, blue indicates no damage) simulated in CabanaPD as a result of vertical tensile loading. Note how the damage pattern reflects the grain boundaries.